Satisfiability implies consistency for the sequent calculus

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author corollary labelled cor:consistency-soundness (2295-2297); the proof span 2299-2311 is the author proof environment that immediately follows it.

Editorial note. The source statement and proof use the unit's iftag wording alternatives; the pinned card keeps the first-order variant, which matches this unit.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/sequent-calculus, counted once: satisfiability implies consistency. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Soundness of the sequent calculus. Every derivable sequent is valid; the unit proves this as its soundness theorem.. It upgrades the derived empty-succedent sequent for Gamma_0 to validity (2303-2305).
- Inconsistency in the sequent calculus. A set Gamma is inconsistent when some finite subset Gamma_0 has a derivation of the sequent Gamma_0 with empty succedent.. It supplies the finite witness set Gamma_0 the proof works with (2301-2303).
- Validity of an empty-succedent sequent. The sequent Gamma_0 with empty succedent is valid exactly when no structure satisfies all sentences of Gamma_0 simultaneously.. It converts the valid sequent into the non-satisfiability of Gamma_0, hence of Gamma (2305-2311).

Source reference. Original theorem thm-sequent-soundness: sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex, source lines 1982–1982. The author proof cites the unit's soundness theorem. The load-bearing work of soundness is carried by that separately proved theorem of the same unit; the counted proof retains its own contrapositive argument in full.

Corollary 0.1. *If* Γ *is satisfiable, then it is consistent.*

Proof. We prove the contrapositive. Suppose that Γ is not consistent. Then there is a finite $\Gamma_0 \subseteq \Gamma$ and !!aderivation of $\Gamma_0 \Rightarrow$. By thm-sequent-soundness, $\Gamma_0 \Rightarrow$ is valid. In other words, for every !!structure $\mathfrak{A}M$, there is $!C \in \Gamma_0$ so that $/ \models M$, and since $\Gamma_0 \subseteq \Gamma$, that $!C$ is also in Γ. Thus, no $\mathfrak{A}M$satisfies Γ, and Γ is not satisfiable. □

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